

Eugene Cheung

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EDUCATION

University of Illinois Urbana-Champaign

Master of Computer Science

Champaign, IL

Aug. 2026 – May 2028

- **Relevant Coursework:** Deep Learning for Healthcare, Artificial Intelligence

University of Washington

Bachelor of Arts in Geography: Data Science

Seattle, WA

Sep. 2022 – Jun. 2026

- **Relevant Coursework:** Machine Learning, Data Structures & Algorithms, Algorithms & Computational Complexity, Database Systems, Probability, Matrix Algebra

EXPERIENCE

Student Researcher

University of Washington - Bothell (DAIS Lab, Dr. Dong Si)

Sep. 2025 – Jun. 2026

Remote

- Built a **NumPy/Biopython** pipeline to extract RNA motifs from molecular structures and cryo-EM maps, curating **31,453 samples** across **24 classes** for model training.
- Trained **PyTorch** RNA motif classifiers and ran **10 controlled ablations** of sugar-atom masking, reducing relative classification error by **~25%** and improving accuracy by **4.38 percentage points**.
- Automated **Python/Bash** inference on a **multi-GPU Linux cluster**, processing **1,995 cryo-EM maps** and completing **200 model evaluations** for downstream RNA structural analysis.

Backend SDE Intern

Customboxes.io

Jun. 2025 – Dec. 2025

Remote

- Built an **OpenAI Agents SDK** packaging-design assistant with **23 function-calling tools** and **SSE** streaming for live preview updates, reducing user design time by **90%**.
- Developed a **Docker/Selenium** pipeline to extract **4 branding asset types**—logos, fonts, links, and QR codes—from merchant websites for automated packaging design.
- Built a **PyTorch** computer vision pipeline combining **CLIP embeddings** and **OCR text anchoring** to remove **70–80% of dieline noise** from packaging PDFs for design analysis.

Software Engineer, Team Lead

Analog Club

Sep. 2024 – Jun. 2026

Seattle, WA

- Led **8 software engineers**, reviewed and merged **20+ PRs**, and deployed a **React SPA** for a community of **873 members** using Agile and Git workflows.
- Built a mobile-first **React** application with **PostgreSQL** and the **Google Calendar API** for discovering club events, digital magazines, and multimedia galleries.

PROJECTS

Mahjong House | TypeScript, Node.js, Python, PyTorch, WebSockets, Docker

Jul. 2026

- Distilled a **PyTorch** policy-value network from **~51k self-play trajectories** and combined it with expected-value search, exporting weights to JSON for **sub-50ms TypeScript inference**.
- Improved the agent's composite evaluation score by **175 points** versus an aggressive baseline in seeded A/B ablations; raised winning-tile enumeration accuracy from **84% to 100%** on a Hugging Face benchmark.
- Deployed the agent in a real-time **Node.js/WebSocket** multiplayer platform with per-player state redaction and live **JSONL** decision logs to support subsequent ML training.

elosurgery | FastAPI, Next.js, Python, TypeScript, D3.js, SQLite

Apr. 2026

- Processed **5,000 games per request** without frontend timeouts using an asynchronous **FastAPI** background worker and **Next.js** client polling.
- Analyzed gameplay and clock-use patterns across **thousands of PGNs** with a **python-chess** pipeline, **Kaplan-Meier** survival curves, and **HHI** concentration measures.
- Implemented versioned **SQLite** caching to eliminate repeated API parsing and enable **near-instant dashboard loads**, with **D3.js** heatmaps and automated narrative summaries.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, Bash

Frameworks: React, Node.js, FastAPI, Next.js, PyTorch, OpenAI Agents SDK, D3.js

Developer Tools: Git, Docker, PostgreSQL, SQLite, WebSockets, Linux

AI/ML Domains: Deep Learning, Computer Vision, Agentic Systems, Game AI